Two-minute cadence for TIC 402710280, a 36.3-day sub-Neptune candidate with a pre-registered transit in Sector 114

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1 DDT Justification

TIC 402710280 (TYC 9041-03677-1, $T = 9.24$) hosts a transiting sub-Neptune candidate, TIC 402710280.01, with $P = 36.33986 \pm 0.00051$ d (Vora, 2026a). *TESS* has observed the star only in full-frame images (FFIs). The period could be fixed only once the Sector 100 and 102 transits (2026 February and March) fell on consecutive cycles, so the target was not proposed in the Cycle 9 Guest Investigator call. *TESS* will observe the star in Sectors 113 and 114 (camera 4), and according to `tess-point` (Burke et al., 2020) it will not observe it again through Sector 134 (2028 September). The transit predicted for 2027 February 19 in Sector 114 is therefore the only transit *TESS* can record for at least the next two years. Without a DDT allocation it will be recorded only in 200-s FFIs, and no target pixel files or SPOC two-minute light curves will exist.

2 Scientific Justification

Four transits, in Sectors 39, 93, 100 and 102, share a depth of 2092 ± 47 ppm and a duration of 5.27 ± 0.05 h (Figure 1). They come from three independent reductions (TESS-SPOC, QLP and our own FFI photometry). Every shorter alias period predicts transits in covered data that are absent at $\geq 18.6\sigma$. With the host's TIC parameters ($T_{\rm eff} = 5182$ K, $R_\star = 0.82\,R_\odot$, $M_\star = 0.88\,M_\odot$) the candidate has $R_p = 3.75 \pm 0.22\,R_\oplus$, $a = 0.206$ AU and $T_{\rm eq} \approx 500$ K. The vetting is as follows:

- Difference-image centroids place the signal $1.5'' \pm 1.3''$ from the target.
- The only *Gaia* star close enough to contribute ($3.5''$ away, 354 times fainter) would need a 74% eclipse, whose shape is disfavoured for any main-sequence pair ($\Delta\chi^2 \geq 142$).
- No secondary eclipse is detected.
- `TRICERATOPS` (Giacalone et al., 2021) gives FPP $= 0.003 \pm 0.004$ and NFPP $< 10^{-6}$. No high-resolution imaging exists, so we treat these as upper bounds on the evidence rather than a validation.

Long-period, temperate sub-Neptunes around stars this bright are uncommon among known planets. The expected radial-velocity semi-amplitude, about 2.8 m s^{-1} for a mass of about 14 $M_\oplus$, is within reach of current spectrographs.

Two-minute data in Sector 114 would address three open questions.

(1) The registered test of the ephemeris. The transit times scatter about a linear ephemeris by 21.8 min rms, against formal uncertainties of 1.7–3.3 min. We find no instrumental cause, and four epochs cannot separate timing systematics from transit-timing variations. We registered the prediction for epoch 57 ($\mathrm{BJD_{TDB}}$ 2461455.978 ± 30 min) and the rules that decide its outcome before the data exist (Vora, 2026b). That protocol names SPOC two-minute light curves as the first-choice data, ahead of any FFI product. Its pixel-level check of the signal's position uses target pixel files.

(2) An independent measurement of the timing. Our reductions produced two of the four mid-times. A SPOC light curve gives a mid-time from photometry that we did not make. If

the timing scatter comes from our light-curve production, it should disappear in SPOC data. If it comes from a perturbing body, it should persist.

(3) The 3.78-day signal. The light curve also contains a 407 ± 26 ppm, 1.3-h signal with $P = 3.78371$ d (Vora, 2026a). It may come from the target, possibly as a grazing transit, or from the $3.5''$ neighbour. Sectors 113 and 114 together contain about a dozen of its events. At two-minute cadence they would resolve its ingress shape, which tests the grazing geometry. The target pixel files would also tighten its difference-image centroid, which is currently $5.2'' \pm 6.7''$ from the target.

3 Target Justification and Feasibility

The request is for one target, TIC 402710280 (RA 248.5236°, Dec −63.5497°), at two-minute cadence in Sectors 113 and 114. Sector 114 has priority 1.0. `tess-point` places the star on silicon in both sectors: camera 4, CCD 4 (column 385, row 209) in Sector 113, and camera 4, CCD 1 (column 646, row 475) in Sector 114. The TIC contamination ratio is 0.06, and there are no stars brighter than 7th magnitude in the field.

The epoch-57 transit is predicted for 2027 February 19 at 11:29 UTC, with $\sigma = 30$ min (34 min including the covariance of T_0 and P). This is about six days after the sector begins, well inside the first orbit. The adjacent transits (2027 January 14 and March 27) fall outside the two sectors.

In FFIs each transit is measured with a depth uncertainty of 86–144 ppm, so a single transit is detected at about 15–24σ. SPOC photometry at $T = 9.2$ should do at least as well. Ingress and egress each last about 14 min, so they span seven two-minute cadences compared with four 200-s cadences. This improves the mid-time and the shape constraint, which is the discriminant between the planet and a diluted eclipse of the neighbour.

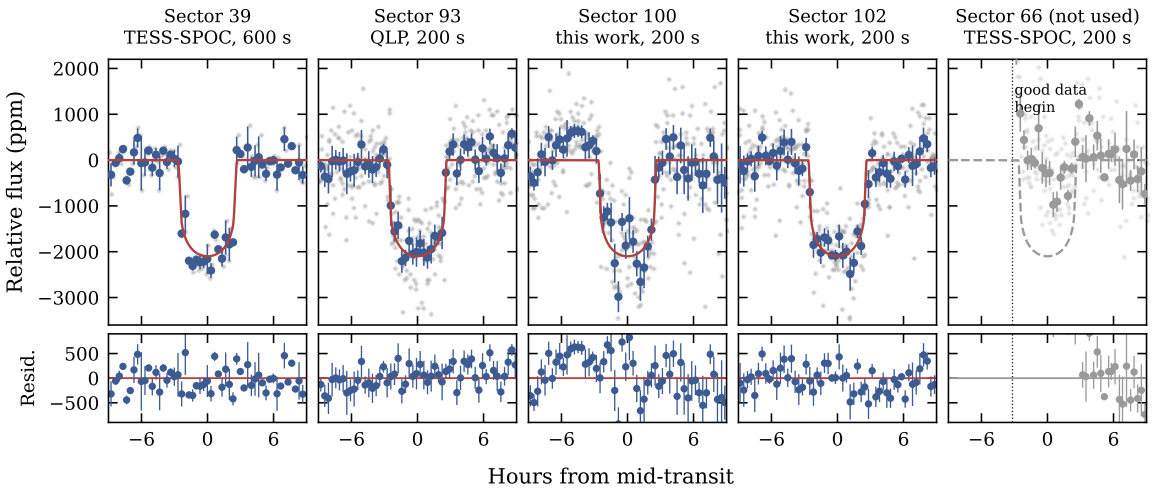


Figure 1: The four transits of TIC 402710280.01 with the median transit model and residuals (20-min bins). The rightmost panel shows the Sector 66 window, which scattered light corrupted and which is excluded from all fits (Vora, 2026a).

References

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